

Platform API overview

The platform module in React Native allows you to detect which operating system your app is running on and allow to write platform specific code.

```
import { Platform } from 'react-native';
```

Key Methods

Platform.OS

Returns a string indicating the operating system

Practical Use Case

```
const topPadding = Platform.OS === 'ios' ? 40 : 20
```

// Different icons

```
const icon = Platform.OS === 'android'
```

```
  ? 'md-checkmark'
```

```
  : 'ios-checkmark';
```


Platform.select()

Allows you to specify different values for different platforms in a cleaner way.

Simple Value Selection

```
const buttonColor = Platform.select({  
  ios: 'blue',  
  android: 'green',  
  default: 'gray'  
});
```

Component Selection

```
const ComponentToRender = Platform.select({  
  ios: () => <IOSComponent/>,  
  android: () => <AndroidComponent/>,  
})();
```

Style Selection

```
const styles = StyleSheet.create({  
  container: {  
    ...Platform.select({  
      ios: {  
        shadowColor: '#000',  
        shadowOffset: { width: 0, height: 2},
```



```
android: {  
  elevation: 4,  
}
```

Platform Version:

Returns the version of operating system

// On Android: returns API level (28, 29, 30)

// On iOS: returns OS version ("14.0")

```
if (Platform.Version >= 28) {
```

// Use feature available in API 28+

```
}
```

Platform-Specific Component overview

File Naming Convention

React Native automatically loads the correct file based on platform using file extensions

Naming Pattern

ComponentName.ios.js → Loaded on iOS

ComponentName.android.js → Loaded on Android

ComponentName.js → Fallback for both

(if no platform-specific file exists)

Platform Specific Components by React Native

Android: Toast Android, BackHandler,
Permission Android, DrawerLayoutAndroid
TimePickerAndroid, DatePickerAndroid

iOS: ActionSheetIOS, Settings

both but styled differently: Button, Switch
StatusBar, ProgressBar
Picker

Some Platform-specific API

Android-specific:

ToastAndroid - Show toast notifications

BackHandler - Handle hardware back button

Permissions Android - Request runtime permissions

iOS-specific:

ActionSheetIOS - Native iOS action sheets

Settings - Access app settings

cross-platform alternatives:

- Use `Alert` instead of `ToastAndroid`
- Use third-party libraries for unified experienciend